

Seminar 5 - Second Order Linear ODEs with Constant Coefficients

$$\boxed{y'' + ay' + by = 0} ; a, b \in C(I), I \subseteq \mathbb{R} \text{ interval}$$

y_1, y_2 solutions $\Rightarrow y_1 + y_2$ solution ; cy_1 solution ($c \in \mathbb{R}$).

The set of solutions is a vector (linear) space of dimension 2.

We need just two linear independent solutions $\{y_1, y_2\}$

$$\boxed{a, b \in \mathbb{R}} \quad y = e^{\lambda x} \Rightarrow \lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + b e^{\lambda x} = 0 \mid : e^{\lambda x} \Rightarrow \boxed{\lambda^2 + a\lambda + b = 0}$$

$$\Delta > 0$$

$$\Delta = 0$$

$$\Delta < 0$$

Case 1: $\Delta > 0 \Rightarrow \exists \lambda_1, \lambda_2 \in \mathbb{R}, \lambda_1 \neq \lambda_2 \Rightarrow y_1 = e^{\lambda_1 x}, y_2 = e^{\lambda_2 x}$

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}, c_1, c_2 \in \mathbb{R}$$

Ex 1: $y'' - 5y' + 6y = 0 ; \lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda_1 = 2, \lambda_2 = 3$
 $y_1 = e^{2x}, y_2 = e^{3x} ; y = c_1 e^{2x} + c_2 e^{3x}, c_1, c_2 \in \mathbb{R}$

Case 2 : $\Delta = 0 \Rightarrow \lambda_1 = \lambda_2$; $y_1 = e^{\lambda_1 x}$; $y_2 = x \cdot e^{\lambda_1 x}$
 please, check that this is a solution. *

Ex 2: $y'' - 4y' + 4y = 0$
 $\lambda^2 - 4\lambda + 4 = 0 \Rightarrow \lambda_1 = \lambda_2 = 2 \Rightarrow y_1 = e^{2x}$; $y_2 = x e^{2x}$
 $y = e^{2x} \cdot (C_1 + x C_2)$, $C_1, C_2 \in \mathbb{R}$

* $\begin{cases} y_2 = x e^{\lambda x} \\ y_2' = e^{\lambda x} + x \cdot \lambda e^{\lambda x} = e^{\lambda x} (1 + \lambda x) \\ y_2'' = \lambda e^{\lambda x} + \lambda e^{\lambda x} (1 + \lambda x) = e^{\lambda x} (\lambda^2 x + 2\lambda) \end{cases} \begin{array}{l} \cdot b \\ \cdot a \\ \cdot 1 \end{array}$

$0 = e^{\lambda x} (\lambda^2 x + 2\lambda + \underline{a} + \underline{a\lambda x} + \underline{bx}) \Leftrightarrow 0 = x \cdot (\underbrace{\lambda^2 + a\lambda + b}_0) + (a + 2\lambda) \Leftrightarrow$

$\Leftrightarrow a + 2\lambda = 0 \Leftrightarrow \lambda = -\frac{a}{2} \Leftrightarrow \Delta = 0.$

Case 3: $\Delta < 0 \Rightarrow \lambda_{1,2} = a \pm ib, b \neq 0$;

Ex 3: $y'' + 2y' + 5y = 0$

$$\Delta = 4 - 20 = -16 ; \sqrt{\Delta} = 4i$$

$$\lambda_{1,2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

$$y_1 = e^{-x} \cos 2x, y_2 = e^{-x} \sin 2x$$

$$y = e^{-x} (C_1 \cos 2x + C_2 \sin 2x)$$

$$e^{\alpha + i\beta} = e^{\alpha} (\cos \beta + i \sin \beta)$$

$$e^{\lambda_1 x} = e^{ax + ibx} = e^{ax} (\cos bx + i \sin bx)$$

$$e^{\lambda_2 x} = e^{ax - ibx} = e^{ax} (\cos bx - i \sin bx)$$

$$\frac{e^{\lambda_1 x} + e^{\lambda_2 x}}{2} = \boxed{e^{ax} \cos bx = y_1}$$

$$\frac{e^{\lambda_1 x} - e^{\lambda_2 x}}{2i} = \boxed{e^{ax} \sin bx = y_2}$$

$$y'' + ay' + by = f \quad (2)$$

$y = y_0 + c_1 y_1 + c_2 y_2$ (y_0 is a solution of (2), $\{y_1, y_2\}$ is a base of solutions for $y'' + ay' + by = 0$)
 $[c_1, c_2 \in \mathbb{R}]$

② How do we find y_0 ?

$$y_0 = \varphi_1 y_1 + \varphi_2 y_2 \quad \{ \varphi_1, \varphi_2 \text{ functions} \} \quad \begin{array}{l} \cdot b \\ \cdot a \\ \cdot 1 \end{array}$$

$$y_0' = \cancel{\varphi_1'} y_1 + \varphi_1 y_1' + \cancel{\varphi_2'} y_2 + \varphi_2 y_2' \quad \begin{array}{l} \cdot a \\ \cdot 1 \end{array}$$

$$y_0'' = \boxed{\varphi_1' y_1'} + \varphi_1 y_1'' + \boxed{\varphi_2' y_2'} + \varphi_2 y_2'' \quad \cdot 1$$

$$f = \underbrace{\varphi_1 (y_1'' + ay_1' + by_1)}_0 + \underbrace{\varphi_2 (y_2'' + ay_2' + by_2)}_0 + \varphi_1' y_1' + \varphi_2' y_2'$$

$$\rightarrow \begin{cases} \varphi_1' y_1 + \varphi_2' y_2 = 0 \\ \varphi_1' y_1' + \varphi_2' y_2' = f \end{cases} \rightarrow \varphi_1' = \dots, \varphi_2' = \dots$$

$$\int dx \Rightarrow \varphi_1 = \dots, \varphi_2 = \dots \quad \textcircled{!}$$